

Construction of lump-type and interaction solutions to an extended $(3 + 1)$ -dimensional Jimbo–Miwa-like equation

Feng-Hua Qi^{*,§,††} and Wen-Xiu Ma^{†,‡,§,¶,||,**,§§},

^{*}*School of Information, Beijing Wuzi University, Beijing 101149, China*

[†]*Department of Mathematics, Zhejiang Normal University,
 Jinhua 321004, Zhejiang, China*

[‡]*Department of Mathematics, King Abdulaziz University, Jeddah, Saudi Arabia*

[§]*Department of Mathematics and Statistics, University of South Florida,
 Tampa, FL 33620-5700, USA*

[¶]*School of Mathematics, South China University of Technology, Guangzhou 510640, China*

^{||}*College of Mathematics and Systems Science,
 Shandong University of Science and Technology, Qingdao 266590, Shandong, China*

^{**}*International Institute for Symmetry Analysis and Mathematical Modelling,
 Department of Mathematical Sciences, North-West University,
 Mafikeng Campus, Mmabatho 2735, South Africa*

^{††}*qifenghua434@163.com*

^{§§}*mawx@cas.usf.edu*

Pan Wang

Sports Business School, Beijing Sport University, Beijing 100084, China

Qi-Xing Qu

*School of Information, University of International Business and Economics,
 Beijing 100029, China*

Received 7 October 2019

Revised 9 November 2019

Accepted 13 November 2019

Published 19 February 2020

We present lump-type solutions and interaction solutions to an extended $(3 + 1)$ -dimensional Jimbo–Miwa-like equation. Three classes of lump-type solutions are obtained by the Hirota bilinear method. Interaction solutions are among lump-type solutions, two kink waves and periodic waves, and between two kink waves and a periodic wave are computed. Dynamical characters of the obtained solutions are graphically exhibited. These wave solutions enrich the dynamical theory of higher-dimensional nonlinear dispersive wave equations.

Keywords: Lump-type solution; interaction solution; kink solution; periodic solution.

^{††,§§}Corresponding authors.

1. Introduction

Nonlinear evolution equations model varieties of physical phenomena.^{1–14} Recently, Wazwaz presented two extended $(3+1)$ -dimensional Jimbo–Miwa equations^{15–28}:

$$u_{xxxx} + 3u_x u_{xy} + 3u_y u_{xx} + 2u_{yt} - 3(u_{xz} + u_{yz} + u_{zz}) = 0, \quad (1)$$

and

$$u_{xxxx} + 3u_x u_{xy} + 3u_y u_{xx} - 3u_{yt} + 2(u_{xt} + u_{yt} + u_{zt}) = 0, \quad (2)$$

which describe certain $(3+1)$ -dimensional waves in plasmas and optics.¹⁵ Here, $u(x, y, z, t)$ is a function of the spatial variables x, y, z and the temporal variable t , and the subscripts denote the corresponding partial derivatives.¹⁶ Exact solitary wave solutions,²⁹ multiple soliton solutions,^{16,30} meromorphic exact solutions,³¹ resonant multi-soliton solutions,³² periodic solitary wave solutions,³³ lump solutions and the lump–kink solutions³⁴ of Eqs. (1) and (2) have been obtained by various methods.³⁵

Under the transformation $u = 2(\ln F)_x$, a Hirota bilinear form for Eq. (1) reads

$$(D_x^3 D_y + 2D_t D_y - 3D_x D_z - 3D_y D_z - 3D_z^2)F \cdot F = 0. \quad (3)$$

Recently, a generalization of the bilinear differential operators has been presented as³⁶

$$D_{p,x_1}^{n_1} \cdots D_{p,x_M}^{n_M} (f \cdot g) = \prod_{i=1}^M \left(\frac{\partial}{\partial x_i} + \alpha \frac{\partial}{\partial x'_i} \right)^{n_i} \times f(x_1, \dots, x_M) g(x'_1, \dots, x'_M) \Big|_{x'_1=x_1, \dots, x'_M=x_M}, \quad (4)$$

in which $n_1, \dots, n_M$ are nonnegative integers, with an integer m , $\alpha^m = (-1)^{r(m)}$, if $m \equiv r(m)$ with $0 \leq r(m) < p$. Taking $p = 3$, we have

$$\alpha_3 = -1, \quad \alpha_3^2 = \alpha_3^3 = 1, \quad \alpha_3^4 = -1, \quad \alpha_3^5 = \alpha_3^6 = 1,$$

and we can generalize the bilinear Eq. (3) into

$$\begin{aligned} & (D_{3,x}^3 D_{3,y} + 2D_{3,t} D_{3,y} - 3D_{3,x} D_{3,z} - 3D_{3,y} D_{3,z} - 3D_{3,z}^2)F \cdot F \\ &= 2(3F_{xx}F_{xy} + 2F_{yt}F - 2F_yF_t - 3F_{xz}F + 3F_xF_z \\ &\quad - 3F_{yz}F + 3F_yF_z - 3F_{zz}F + 3F_z^2) = 0, \end{aligned} \quad (5)$$

which is equivalently linked with an extended $(3+1)$ -dimensional Jimbo–Miwa-like equation

$$\begin{aligned} & 2u_{yt} + \frac{3}{8}u^3u_y + \frac{9}{4}uu_yu_x + \frac{3}{4}u^2u_{xy} + \frac{3}{2}u_xu_{xy} + \frac{3}{2}u_yu_{xx} \\ & - 3(u_{xz} + u_{yz} + u_{zz}) + \frac{9}{8}vu^2u_x + \frac{3}{4}vu_x^2 + \frac{3}{4}vu u_{xx} = 0, \end{aligned} \quad (6)$$

where $u_y = v_x$.

In this paper, we will consider Eq. (6). This paper will be structured as follows. In Sec. 2, we will present three classes of lump-type solutions of Eq. (6) through a generalized Hirota bilinear method. In Sec. 3, we will construct and analyze some interaction solutions. In Sec. 4, dynamical properties of the above presented solutions will be graphically discussed. Conclusions will be provided in Sec. 5.

2. Lump-Type Solutions

To search for lump-type solutions of Eq. (6) (see, for example, Refs. 37–45 for other examples), we begin by assuming

$$\begin{cases} F = G^2 + H^2 + a_{11}, & G = a_1x + a_2y + a_3z + a_4t + a_5, \\ H = a_6x + a_7y + a_8z + a_9t + a_{10}, \end{cases} \quad (7)$$

where $a_i (1 \leq i \leq 11)$ are real parameters to be determined. Substituting such a function F in Eqs. (7) into the bilinear Eq. (5), we can, with Maple symbolic computation, get

Case 1.

$$\begin{cases} a_4 = \frac{3(a_3a_2^2 + a_3^2a_2 - a_8^2a_2 + a_1a_3a_2 - a_6a_8a_2 + a_3a_7^2 + a_3a_6a_7 + a_1a_7a_8 + 2a_3a_7a_8)}{2(a_2^2 + a_7^2)}, \\ a_9 = \frac{3(a_8a_2^2 + a_3a_6a_2 + a_1a_8a_2 + 2a_3a_8a_2 + a_7a_8^2 - a_3^2a_7 - a_1a_3a_7 + a_7^2a_8 + a_6a_7a_8)}{2(a_2^2 + a_7^2)}, \\ a_{11} = \frac{(a_1^2 + a_6^2)(a_1a_2 + a_6a_7)(a_2^2 + a_7^2)}{(a_3a_7 - a_2a_8)[(a_1 + a_3)a_7 - a_2(a_6 + a_8)]}, \end{cases} \quad (8)$$

where the involved constants should satisfy $(a_2^2 + a_7^2)(a_3a_7 - a_2a_8)[(a_1 + a_3)a_7 - a_2(a_6 + a_8)] \neq 0$.

Case 2.

$$\begin{aligned} a_2 &= \frac{a_3a_7}{a_8}, & a_4 &= \frac{3(a_3a_7 + a_1a_8 + a_3a_8)}{2a_7}, \\ a_8 &= -\frac{a_1a_3}{a_6}, & a_9 &= \frac{3a_8(a_6 + a_7 + a_8)}{2a_7}, \end{aligned} \quad (9)$$

where $a_6a_7a_8 \neq 0$.

Case 3.

$$\begin{cases} a_2 = \frac{a_3^2 a_7 - a_7 a_8^2}{2a_3 a_8}, \\ a_4 = \frac{3a_3(2a_8^3 + 2a_6 a_8^2 + a_7 a_8^2 + 2a_3^2 a_8 + 2a_1 a_3 a_8 + a_3^2 a_7)}{2a_7(a_3^2 + a_8^2)}, \\ a_9 = \frac{3a_8(2a_6 a_3^2 + a_7 a_3^2 - 2a_1 a_8 a_3 + a_7 a_8^2)}{2a_7(a_3^2 + a_8^2)}, \\ a_{11} = \frac{(a_1^2 + a_6^2)a_7(a_3^2 + a_8^2)(a_1 a_3^2 + 2a_6 a_8 a_3 - a_1 a_8^2)}{2a_3 a_8^2(a_8^3 + a_6 a_8^2 + a_3^2 a_8 + 2a_1 a_3 a_8 - a_3^2 a_6)}, \end{cases} \quad (10)$$

where $a_3 a_7 a_8(a_8^3 + a_6 a_8^2 + a_3^2 a_8 + 2a_1 a_3 a_8 - a_3^2 a_6) \neq 0$.

All the above cases of parameter selections result in three classes of solutions to Eq. (7); and then through the transformation $u = 2(\ln F)_x$ we get three classes of lump-type solutions of Eq. (6), which can be presented as

$$u = \frac{4a_1(a_4 t + a_1 x + a_2 y + a_3 z + a_5) + 4a_6(a_9 t + a_6 x + a_7 y + a_8 z + a_{10})}{(a_4 t + a_1 x + a_2 y + a_3 z + a_5)^2 + (a_9 t + a_6 x + a_7 y + a_8 z + a_{10})^2 + a_{11}}, \quad (11)$$

in which it is sufficient for u to be analytic if $a_{11} > 0$. For example, choosing exact values of parameters for the three cases above, we can get three kinds of lump-type solutions separately.

- (1) Taking $a_1 = 1, a_2 = 2, a_3 = 1, a_5 = 1, a_6 = 1, a_7 = 1, a_8 = 1$ and $a_{10} = 1$ for Case 1, we can get

$$u = \frac{2(4x + 6y + 4z + \frac{66t}{5} + 4)}{(x + y + z + \frac{39t}{10} + 1)^2 + (x + 2y + z + \frac{27t}{10} + 1)^2 + 15}; \quad (12)$$

- (2) Taking $a_1 = 1, a_3 = 1, a_5 = 1, a_6 = 1, a_7 = 1, a_{10} = 1$ and $a_{11} = 1$ for Case 2, we can get

$$u = \frac{8x - 12t + 8}{(x + y - z - \frac{3t}{2} + 1)^2 + (x - y + z - \frac{3t}{2} + 1)^2 + 15}; \quad (13)$$

- (3) Taking $a_1 = 1, a_3 = 1, a_5 = 1, a_6 = 1, a_7 = 1, a_8 = 1$ and $a_{10} = 1$ for Case 3, we can get

$$u = \frac{4(2x + y + 2z + 9t + 2)}{(x + y + z + \frac{3t}{2} + 1)^2 + (x + z + \frac{15t}{2} + 1)^2 + 1}. \quad (14)$$

3. Interaction Solutions

First, in order to construct interaction solutions (see, for example, Refs. 41, 46, 47 and Refs. 48–52 for other examples of nonlinear PDEs and linear PDEs,

respectively), we assume that

$$\begin{cases} F = G^2 + H^2 + L + b_{11}, & G = b_1x + b_2y + b_3z + b_4t + b_5, \\ H = b_6x + b_7y + b_8z + b_9t + b_{10}, \\ L = v_1e^{k_1x+k_2y+k_3z+k_4t+k_5} + v_2e^{-k_1x-k_2y-k_3z-k_4t-k_5} \\ \quad + v_3 \cos(k_6x + k_7y + k_8z + k_9t + k_{10}), \end{cases} \quad (15)$$

where $b_i (1 \leq i \leq 11)$, $k_j (1 \leq j \leq 10)$, v_1 , v_2 and v_3 are all real parameters to be determined. Substituting such a function F by Eqs. (15) into the bilinear Eq. (5), we can, through Maple, obtain

Case 1.

$$\begin{cases} b_4 = \frac{3b_3(b_1 + b_2 + b_3)}{2b_2}, & b_6 = -\frac{b_1b_2}{b_7}, & b_8 = \frac{b_3b_7^2 + b_1(b_2^2 + b_7^2)}{b_2b_7}, \\ k_1 = 0, & k_3 = \frac{(b_1 + b_3)k_2}{b_2}, \\ b_9 = \frac{3(b_2^2 + b_7^2)b_1^2 + 3(b_3^2 + b_3b_2^2 + b_7^2b_2 + 2b_3b_7^2)b_1 + 3b_3(b_2 + b_3)b_7^2}{2b_2^2b_7}, \\ k_8 = \frac{(b_1 + b_3)k_7}{b_2}, & k_4 = \frac{3(b_1 + b_3)(b_1 + b_2 + b_3)k_2}{2b_2^2}, \\ k_6 = 0, & k_9 = \frac{3(b_1 + b_3)(b_1 + b_2 + b_3)k_7}{2b_2^2}, \end{cases} \quad (16)$$

which needs to satisfy $b_2b_7 \neq 0$, to make F to be analytic.

Case 2.

$$\begin{cases} b_2 = 0, & b_3 = -b_1, & b_4 = \frac{3}{2}k_1(k_6^2 - 1), & b_7 = 0, & b_8 = -b_6, \\ b_9 = -\frac{3b_6}{2}, & k_2 = 0, \\ k_3 = -k_1, & k_4 = \frac{3}{2}k_1(k_6^2 - 1), & k_7 = 0, & k_8 = -k_6, & b_9 = -\frac{3k_6}{2}. \end{cases} \quad (17)$$

With $u = 2(\ln F)_x$, a kind of interaction solutions of Eq. (6) can be presented as

$$\begin{aligned} u = & [4b_1(b_1x + b_2y + b_3z + b_4t + b_5) + 4b_6(b_6x + b_7y + b_8z + b_9t + b_{10}) \\ & + 2v_1k_1e^{k_1x+k_2y+k_3z+k_4t+k_5} - 2k_1v_2e^{-k_1x-k_2y-k_3z-k_4t-k_5} \\ & - k_6v_3 \sin(k_6x + k_7y + k_8z + k_9t + k_{10})] \times [(b_1x + b_2y + b_3z + b_4t + b_5)^2 \\ & + (b_9t + b_6x + b_7y + b_8z + b_{10})^2 + b_{11} + v_1e^{k_1x+k_2y+k_3z+k_4t+k_5} \\ & + v_2e^{-k_1x-k_2y-k_3z-k_4t-k_5} + v_3 \cos(k_7y + k_8z + k_9t + k_{10})]^{-1}. \end{aligned} \quad (18)$$

Taking Case 2 as an example, by selecting the parameters of Eq. (18) specially as: $b_3 = 1, b_5 = 1, b_6 = 1, b_{10} = 1, b_{11} = 1, k_1 = 1, k_5 = 1, k_6 = 1$ and $k_{10} = 1$, we can obtain the interaction solutions among the lump, kink wave and periodic wave

$$u = \frac{2[-v_3 \sin(x - z - \frac{3t}{2} + 1) - 3t + v_1 e^{x-z+1} - v_2 e^{-x+z-1} + 4x - 4z + 4]}{v_3 \cos(x - z - \frac{3t}{2} + 1) + (x - z - \frac{3t}{2} + 1)^2 + v_1 e^{x-z+1} + v_2 e^{-x+z-1} + (x - z + 1)^2 + 1}, \quad (19)$$

and if further assuming

- (1) $v_1 = 1, v_2 = 1$ and $v_3 = 0$, we can obtain the interaction solution between the lump and two kink waves of Eq. (6)

$$u = \frac{2(e^{x-z+1} - e^{-x+z-1} + 4x - 4z - 3t + 4)}{(x - z - \frac{3t}{2} + 1)^2 + (x - z + 1)^2 + e^{x-z+1} + e^{-x+z-1} + 1}; \quad (20)$$

- (2) $v_1 = 1, v_2 = 0$ and $v_3 = 0$, we can obtain the mixed lump-kink wave solution of Eq. (6)

$$u = \frac{2(e^{x-z+1} + 4x - 4z - 3t + 4)}{(x - z - \frac{3t}{2} + 1)^2 + (x - z + 1)^2 + e^{x-z+1} + 1}; \quad (21)$$

- (3) $v_1 = 0, v_2 = 0$ and $v_3 = 1$, we can obtain the interaction solution between a lump and a periodic wave of Eq. (6)

$$u = \frac{2[-\sin(x - z - \frac{3t}{2} + 1) + 4x - 4z - 3t + 4]}{(x - z - \frac{3t}{2} + 1)^2 + \cos(x - z - \frac{3t}{2} + 1) + (x - z + 1)^2 + 1}. \quad (22)$$

Second, for the purpose of constructing periodic wave solutions,^{19,23,53,54} we assume

$$F = m_1 e^{l_1 x + l_2 y + l_3 z + l_4 t + l_5} + m_2 e^{-l_1 x - l_2 y - l_3 z - l_4 t - l_5} + m_3 \cos(l_6 x + l_7 y + l_8 z + l_9 t + l_{10}), \quad (23)$$

in which $l_j (1 \leq j \leq 10)$, m_1 , m_2 and m_3 are all real parameters to be determined. Substituting such a function F by Eq. (23) into the bilinear Eq. (5), we can, through Maple, obtain

$$l_2 = 0, \quad l_3 = -l_1, \quad l_7 = 0, \quad l_8 = -l_6, \quad (24)$$

and then through the transformation $u = 2(\ln F)_x$ we get periodic wave solutions of Eq. (6)

$$u = \frac{2[l_1 m_1 e^{l_1 x - l_1 z + l_4 t + l_5} - l_1 m_2 e^{-l_1 x + l_1 z - l_4 t - l_5} - l_6 m_3 \sin(l_6 x - l_6 z + l_9 t + l_{10})]}{m_1 e^{l_1 x - l_1 z + l_4 t + l_5} + m_2 e^{-l_1 x + l_1 z - l_4 t - l_5} + m_3 \cos(l_6 x - l_6 z + l_9 t + l_{10})}. \quad (25)$$

4. Dynamical Features

In the following, Figs. 1–4 depict dynamic features and energy distributions of the lump-type solutions, interaction solutions and periodic wave solutions. Figure 1 illustrates three-dimensional plots of three types of lump-type waves with Eqs. (12)–(14) separately. Figures 2 and 3 show the different interaction waves with different parameters. Characters of the interaction process among lump, kink wave and periodic wave are dependent on the rational quadratic function, exponential function (including hyperbolic function) and triangle function.⁵¹ Figure 2 depicts interactions among the lump, kink wave and periodic wave. If we take $v_3 = 1$ (see Fig. 2(a)), the wave has one peak and one valley, when we increase the value of v_3

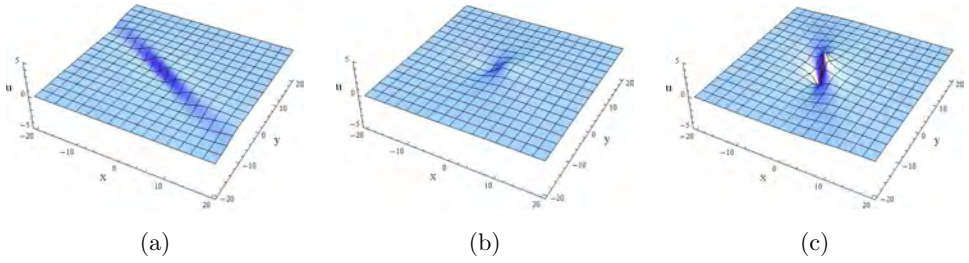


Fig. 1. (Color online) Lump-type solutions of (a) Eq. (12), (b) Eq. (13) and (c) Eq. (14).

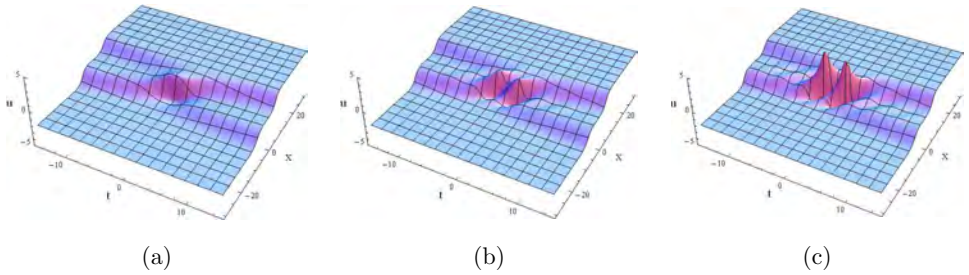


Fig. 2. (Color online) Interaction solutions of Eq. (19) with $v_1 = v_2 = 1$: (a) $v_3 = 1$, (b) $v_3 = 5$ and (c) $v_3 = 10$.

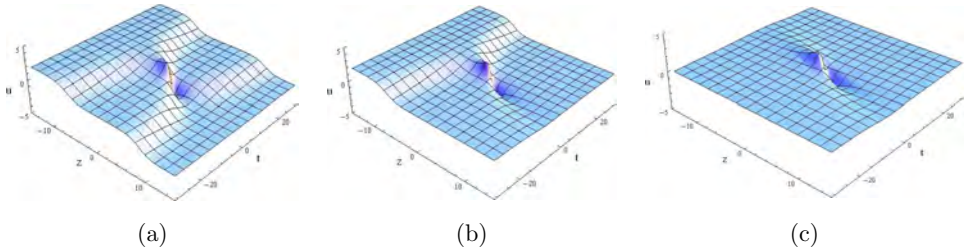


Fig. 3. (Color online) Interaction solutions between: (a) a lump and two kink waves of Eq. (20), (b) a mixed lump–kink wave of Eq. (21) and (c) a lump and a periodic wave of Eq. (22).

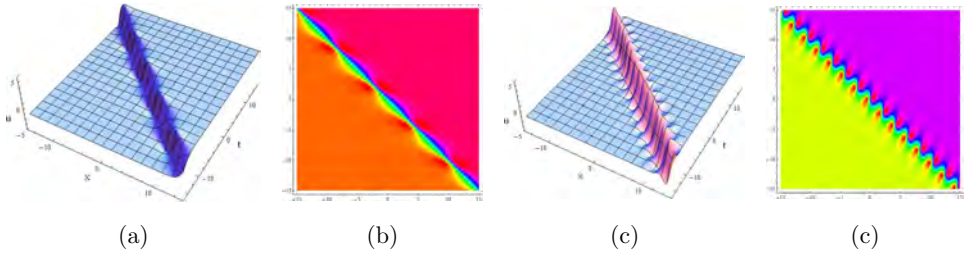


Fig. 4. (Color online) Periodic wave solutions (25) with $l_1 = 1, l_4 = 1, l_5 = 1, l_9 = 1, m_1 = 1, m_2 = 1, m_3 = 1$: (a) $l_6 = 0$; (b) is the corresponding contour plot of (a); (c) $l_6 = 2$; (d) is the corresponding contour plot of (c).

(by taking $v_3 = 5$ and $v_3 = 10$), it splits into two peaks and two valleys gradually with the increase of the amplitudes of the peaks and valleys (see Figs. 2(b) and 2(c)). Figure 3(a) is an interaction between a lump and two kink wave of Eq. (20). Figure 3(b) is a lump–kink wave based on Eq. (21). Figure 3(c) is an interaction wave between a lump and a periodic wave of Eq. (22). Figure 4 shows periodic waves (25) with $l_1 = 1, l_4 = 1, l_5 = 1, l_9 = 1, m_1 = 1, m_2 = 1, m_3 = 1$ along with different value of l_6 , Fig. 4(a) is with $l_6 = 0$ and Fig. 4(c) with $l_6 = 2$, Figs. 4(b) and 4(d) are corresponding contour plots of Figs. 4(a) and 4(c), respectively.

5. Conclusions

In this paper, we have considered the construction of lump-type and interaction solutions to an extended $(3 + 1)$ -dimensional Jimbo–Miwa-like equation, which has potential applications in engineering and physical sciences. All the computations are based on a generalized Hirota bilinear form. Three classes of lump-type solutions have been obtained. Interaction solutions are among lump-type solutions, two kink waves and periodic waves, and between two kink waves and periodic waves. Dynamical features and energy distributions of the presented solutions have been depicted through various plots. Such wave solutions are expected to have applications in modeling nonlinear wave phenomena in nature.

Acknowledgments

This work has been supported by the National Natural Science Foundation of China under Grant Nos. 11605011, 71501015, 11801597, 71801047, 11301454, 11301331, 11371086, 11571079 and 51771083, the State Scholarship Fund of China, Beijing Social Science Foundation (No. 17GLC066), Beijing Municipal University's High-level Innovation Team Construction Project (IDHT20180510), the National Science Foundation (DMS-1664561), the Beijing Philosophy and Social Sciences Planning Project (No. 17GLC052), UIBE Excellent Young Scholar project (No. 18YQ12), Foundation for Disciplinary Development of SIT in UIBE, and Fundamental Research Funds for the Central Universities in UIBE (CXTD10-06).

References

1. W. X. Ma, *Phys. Lett. A* **379** (2015) 1975.
2. W. X. Ma and Y. Zhou, *J. Differ. Equ.* **264** (2018) 2633.
3. T. Xu, S. Lan, M. Li, L. L. Li and G. W. Zhang, *Physica D* **390** (2019) 47.
4. M. Li, J. J. Shui and T. Xu, *Appl. Math. Lett.* **83** (2018) 110.
5. M. Li, T. Xu and D. X. Meng, *J. Phys. Soc. Jpn.* **85** (2016) 124001.
6. R. Guo, H. Q. Hao and L. L. Zhang, *Nonlinear Dynam.* **74** (2013) 701.
7. Z. J. Yang, S. M. Zhang, X. L. Li and Z. G. Pang, *Appl. Math. Lett.* **82** (2018) 64.
8. X. J. Zhao, R. Guo and H. Q. Hao, *Appl. Math. Lett.* **75** (2018) 114.
9. M. Li and T. Xu, *Phys. Rev. E* **91** (2015) 033202.
10. Y. Jiang and B. Tian, *Superlattices Microstruct.* **111** (2017) 350.
11. Y. Jiang, B. Tian, K. Sun and L. C. Liu, *Appl. Math. Lett.* **53** (2016) 69.
12. H. Q. Zhang, M. Y. Zhang and R. Hu, *Appl. Math. Lett.* **76** (2018) 170.
13. H. Q. Zhang and F. Chen, *Appl. Math. Lett.* **88** (2019) 237.
14. H. Q. Zhang, R. Hu and M. Y. Zhang, *Appl. Math. Lett.* **69** (2017) 101.
15. M. Jimbo and T. Miwa, *Publ. Res. Inst. Math. Sci.* **19** (1983) 943.
16. A. M. Wazwaz, *Appl. Math. Lett.* **64** (2017) 21.
17. J. Y. Yang and W. X. Ma, *Comput. Math. Appl.* **73** (2017) 220.
18. W. X. Ma, *Int. J. Nonlinear Sci. Numer. Simul.* **17** (2016) 355.
19. W. Tan, Z. D. Dai, J. L. Xie and L. L. Hu, *Z. Naturforsch.* **73A** (2017) 43.
20. W. X. Ma and J. H. Lee, *Chaos Soliton. Fract.* **42** (2009) 1356.
21. Y. N. Tang, W. X. Ma, W. Xu and L. Gao, *Appl. Math. Comput.* **217** (2011) 21.
22. M. Singh and R. K. Gupta, *Commun. Nonlinear Sci. Numer. Simul.* **37** (2016) 362.
23. Z. D. Dai, J. Liu, X. P. Zheng and Z. J. Liu, *Phys. Lett. A* **372** (2008) 5984.
24. S. Batwa and W. X. Ma, *Comput. Math. Appl.* **76** (2018) 1576.
25. W. X. Ma, Z. Y. Qin and X. Lü, *Nonlinear Dynam.* **84** (2016) 923.
26. J. Y. Yang and W. X. Ma, *Int. J. Mod. Phys. B* **30** (2016) 1640028.
27. X. L. Yong, X. J. Li and Y. H. Huang, *Appl. Math. Lett.* **86** (2018) 222.
28. H. N. Xu, W. Y. Ruan, Y. Zhang and X. Lü, *Appl. Math. Lett.* **99** (2020) 105976.
29. K. K. Ali, R. I. Nuruddeen and A. R. Hadhoud, *Results Phys.* **9** (2018) 12.
30. L. Kaur and A. M. Wazwaz, *Math. Meth. Appl. Sci.* **41** (2018) 7566.
31. H. Li and Y. Z. Li, *Appl. Math. Comput.* **333** (2018) 369.
32. C. K. Kuo and B. Ghanbari, *Nonlinear Dynam.* **1** (2019) 6.
33. J. Manafian, *Comput. Math. Appl.* **76** (2018) 1246.
34. H. Q. Sun and A. H. Chen, *Appl. Math. Lett.* **68** (2017) 55.
35. Y. F. Yue, L. L. Huang and Y. Chen, *Appl. Math. Lett.* **89** (2019) 70.
36. W. X. Ma, *Stud. Nonlinear Sci.* **2** (2011) 140.
37. X. Lü and W. X. Ma, *Nonlinear Dynam.* **85** (2016) 1217.
38. J. G. Liu, *Appl. Math. Lett.* **86** (2018) 36.
39. X. Lü, S. T. Chen and W. X. Ma, *Nonlinear Dynam.* **86** (2016) 523.
40. H. Q. Zhang and W. X. Ma, *Nonlinear Dynam.* **87** (2017) 2305.
41. W. X. Ma, X. L. Yong and H. Q. Zhang, *Comput. Math. Appl.* **75** (2018) 289.
42. S. T. Chen and W. X. Ma, *Front. Math. China* **13** (2018) 525.
43. W. X. Ma, J. Li and C. M. Khalique, *Complexity* **2018** (2018) 9059858.
44. W. X. Ma, *J. Appl. Anal. Comput.* **9** (2019).
45. L. N. Gao, Y. Y. Zi, Y. H. Yin, W. X. Ma and X. Lü, *Nonlinear Dynam.* **89** (2017) 2233.
46. X. Lü, W. X. Ma and C. M. Khalique, *Appl. Math. Lett.* **50** (2015) 37.
47. X. Lü, W. X. Ma, J. Yu and C. M. Khalique, *Commun. Nonlinear Sci. Numer. Simul.* **31** (2016) 40.

- 48. W. X. Ma, *J. Geom. Phys.* **133** (2018) 10.
- 49. W. X. Ma, *East Asian J. Appl. Math.* **9** (2019) 185.
- 50. W. X. Ma, *Acta Math. Sci.* **39B** (2019) 498.
- 51. C. H. He, Y. N. Tang and J. L. Ma, *Comput. Math. Appl.* **76** (2018) 2141.
- 52. Y. F. Hua, B. L. Guo, W. X. Ma and X. Lü, *Appl. Math. Model.* **74** (2019) 184.
- 53. S. L. Zhang, *SN Appl. Sci.* **1** (2019) 201.
- 54. X. X. Du, B. Tian and Y. Yin, *Waves Random Complex Media* (2019), doi:10.1080/17455030.2019.1566681.